01_Introduction

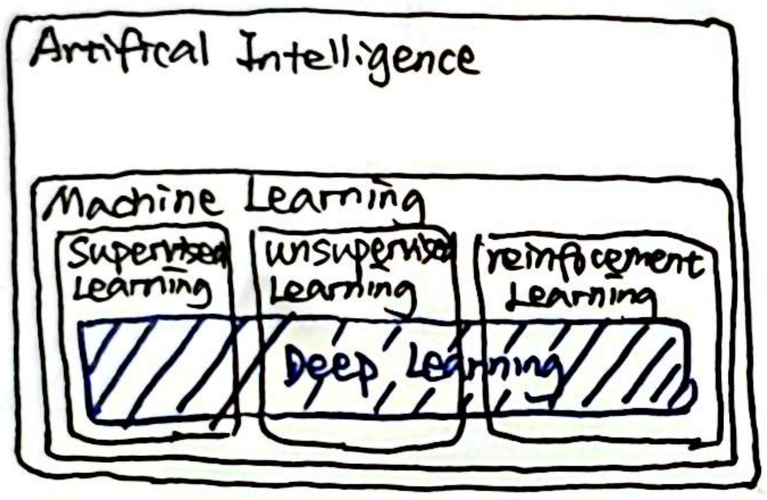
Artifical Intelligence (AI) is concerned with building system that simulate itelligent behavior

- Machine Learning is a subset of AI that learns to make decisions by fitting mathematical models to observed data.

A deep neural network is a type of machine learning model, and the process of fitting these models to data is referred as <u>deep learning</u>. (natural langnage processing, computer vision, speech recognition)

Machine learning methods can coarsely be divided into three areas: (1) supervised (2) unsupervised and (3) reinforcement learning.

Ex. figure:



Machine Learning model is a equation relating Input and output. To train or fit a model, we search through a family of possible equations to find one that fits training data well.

02_Supervised Learning (监督学习)

Model: $y = f[x, \phi]$,

ϕ is the parameter of the function f.

- Training dataset: $\{x_i, y_i\}_{i=1}^{I}$
- Loss (cost) Function:

  $L[\phi, f[x, \phi], \{x_i, y_i\}_{i=1}^{I}]$

  for short. $L[\phi]$
- test dataset: didn't observed during training

Ex: 1-D Linear regression:

$$y = f[x, \phi] = \phi_0 + \phi_1 x$$

ϕ_0: offset, ϕ_1: slope.

$$L[\phi] = \sum_{i=1}^{I} (f[x_i, \phi] - y_i)^2$$

. Least-squares loss function

Goal: $\hat{\phi} = \arg\min_{\phi} [L[\phi]]$

$$= \arg\min_{\phi} \left[\frac{1}{I}\sum_{i=1}^{I} (f[x_i, \phi] - y_i)^2\right]$$

To solve it:

To get cost function

$$L[\phi] = \sum_{i=1}^{I} (\phi_0 + \phi_1 x_i - y_i)^2$$

$$= \sum_{i=1}^{I} \phi_0^2 + 2\phi_0(\phi_1 x_i - y_i) + (\phi_1 x_i - y_i)^2 \quad ①$$

$$= \sum_{i=1}^{I} (\phi_1 x_i)^2 + 2\phi_1 x_i(\phi_0 - y_i) + (\phi_0 - y_i)^2 \quad ②$$

(① = ②)

- $\frac{\partial L}{\partial \phi_1} = \sum_{i=1}^{I} \left(2x_i^2\phi_1 + 2(\phi_0 - y_i)x_i\right) = 0$
- $\frac{\partial L}{\partial \phi_0} = \sum_{i=1}^{I} 2\phi_0 + 2(\phi_1 x_i - y_i) = 0$

we get

$$2\sum_{i=1}^{I} x_i(x_i\phi_1 + (\phi_0 - y_i)) = 0 \quad ②$$

$$2\sum_{i=1}^{I} (\phi_0 + \phi_1 x_i - y_i) = 0 \quad ①$$

$$\Rightarrow ① = I\cdot\phi_0 + \sum_{i=1}^{I} x_i \cdot \phi_1 - \sum_{i=1}^{I} y_i = 0$$

$$② = \sum_{i=1}^{I} x_i^2 \cdot \phi_1 + \sum_{i=1}^{I} x_i \cdot \phi_0 - \sum_{i=1}^{I} x_i y_i = 0$$

By ~~x_i ① - ②~~

$$\hat{\phi}_0 = \bar{y} - \hat{\phi}_1 \bar{x}$$

$$\hat{\phi}_1 = \frac{\sum_{i=1}^{I} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{I} (x_i - \bar{x})^2}$$